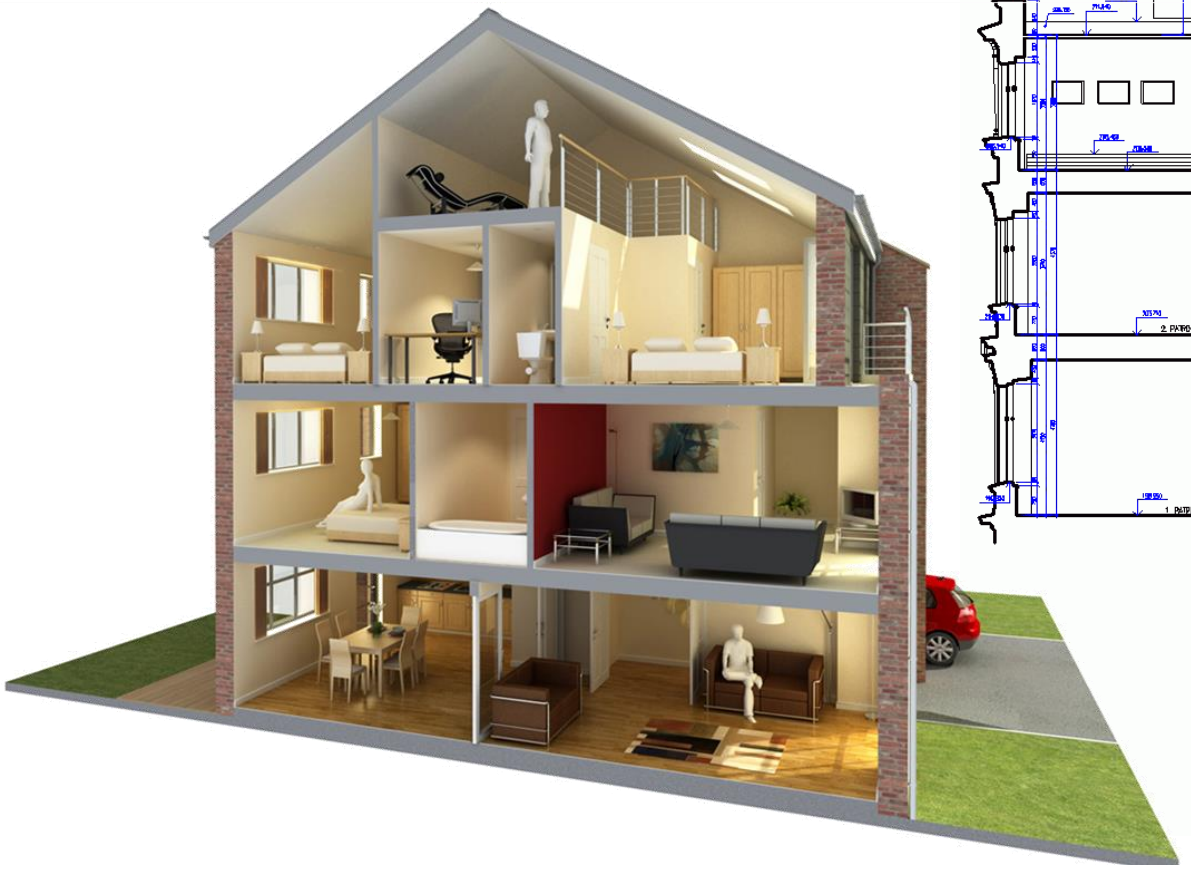
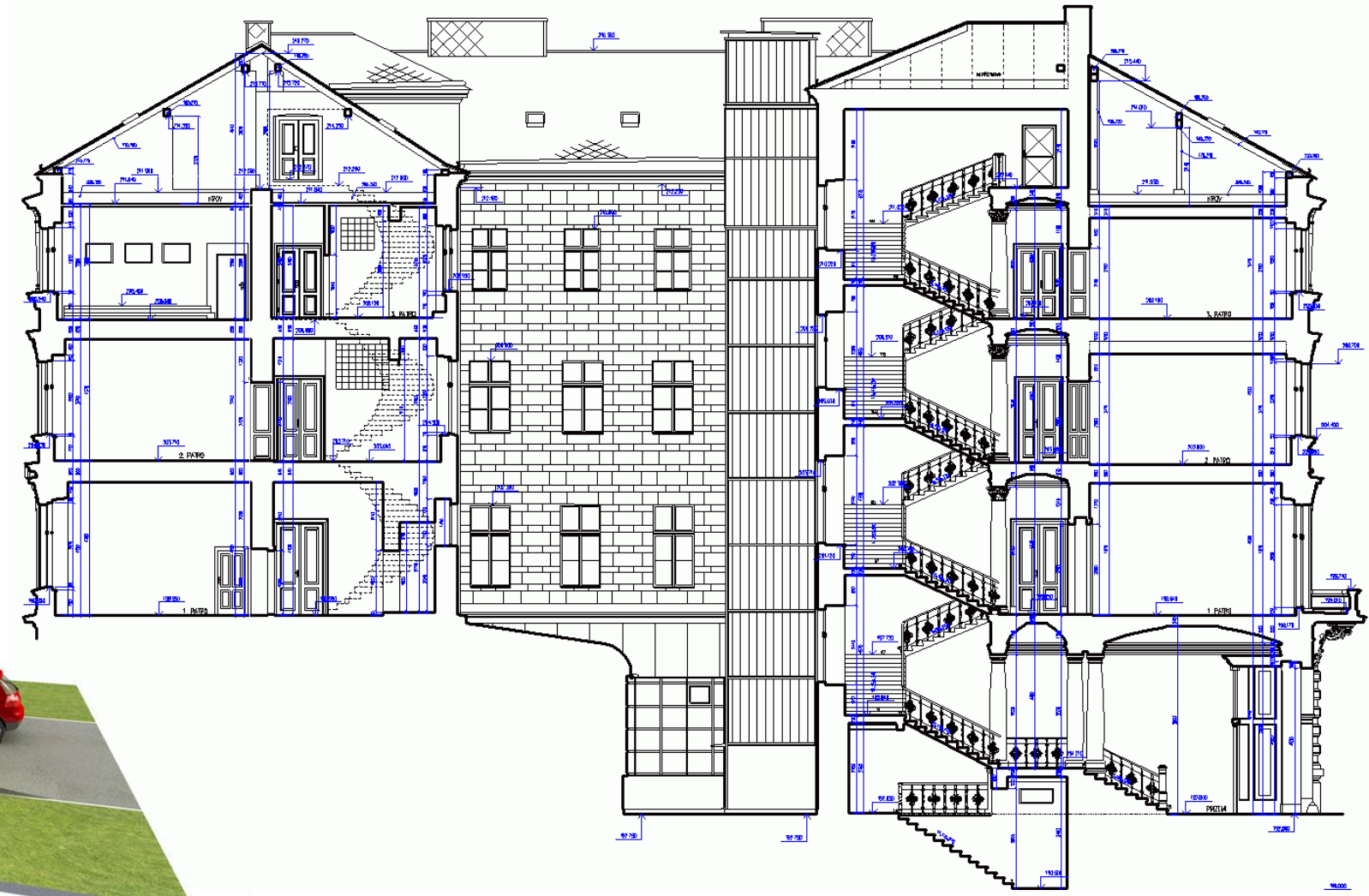
Components of building’s

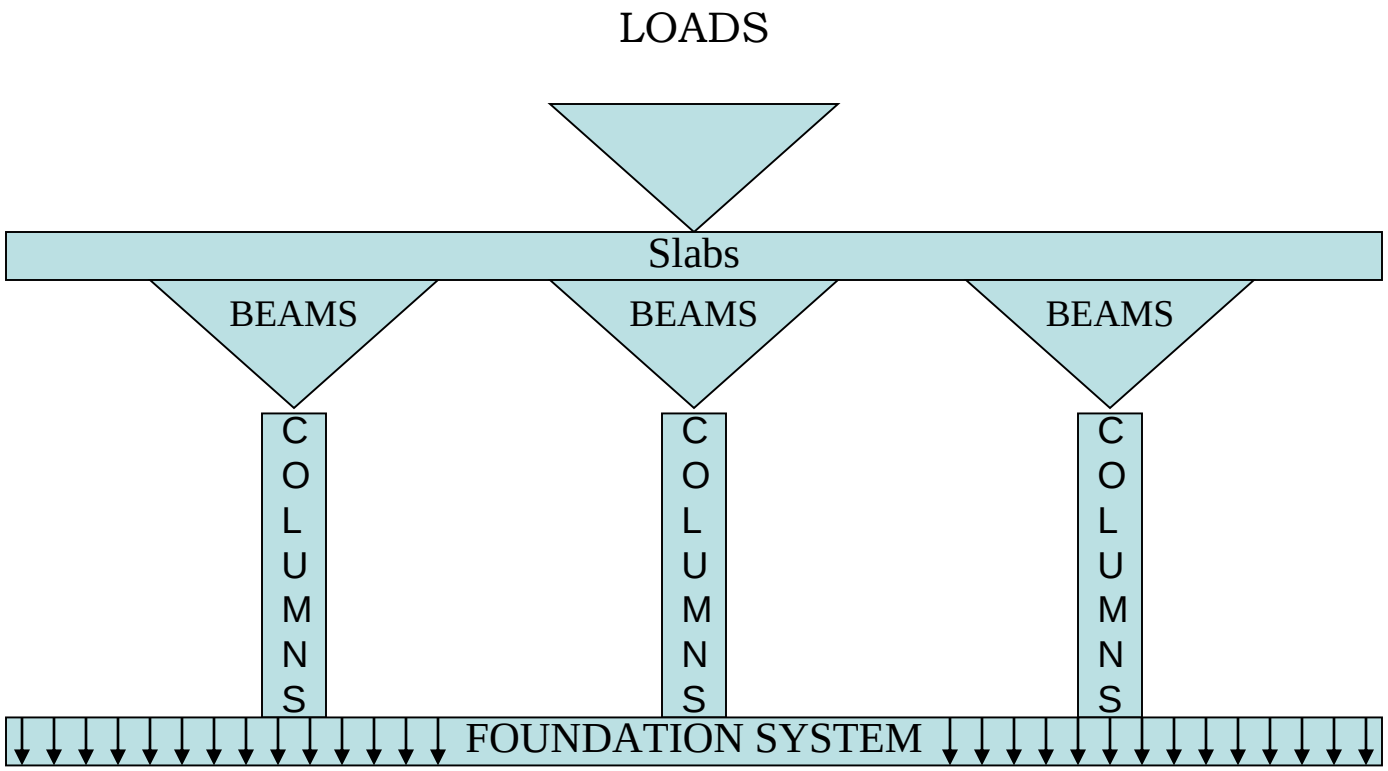




What is a structure?

- A structure is a system of inter connected elements to carry loads safely to under ground earth.
- Basically in building structures there are 2 types of structures:
  - (1) Framed structure
  - (2) Load bearing structure.

Transmission of Loads



THEY MADE IT! THE LOADS HAVE REACHED THE GROUND!

Loads to Consider

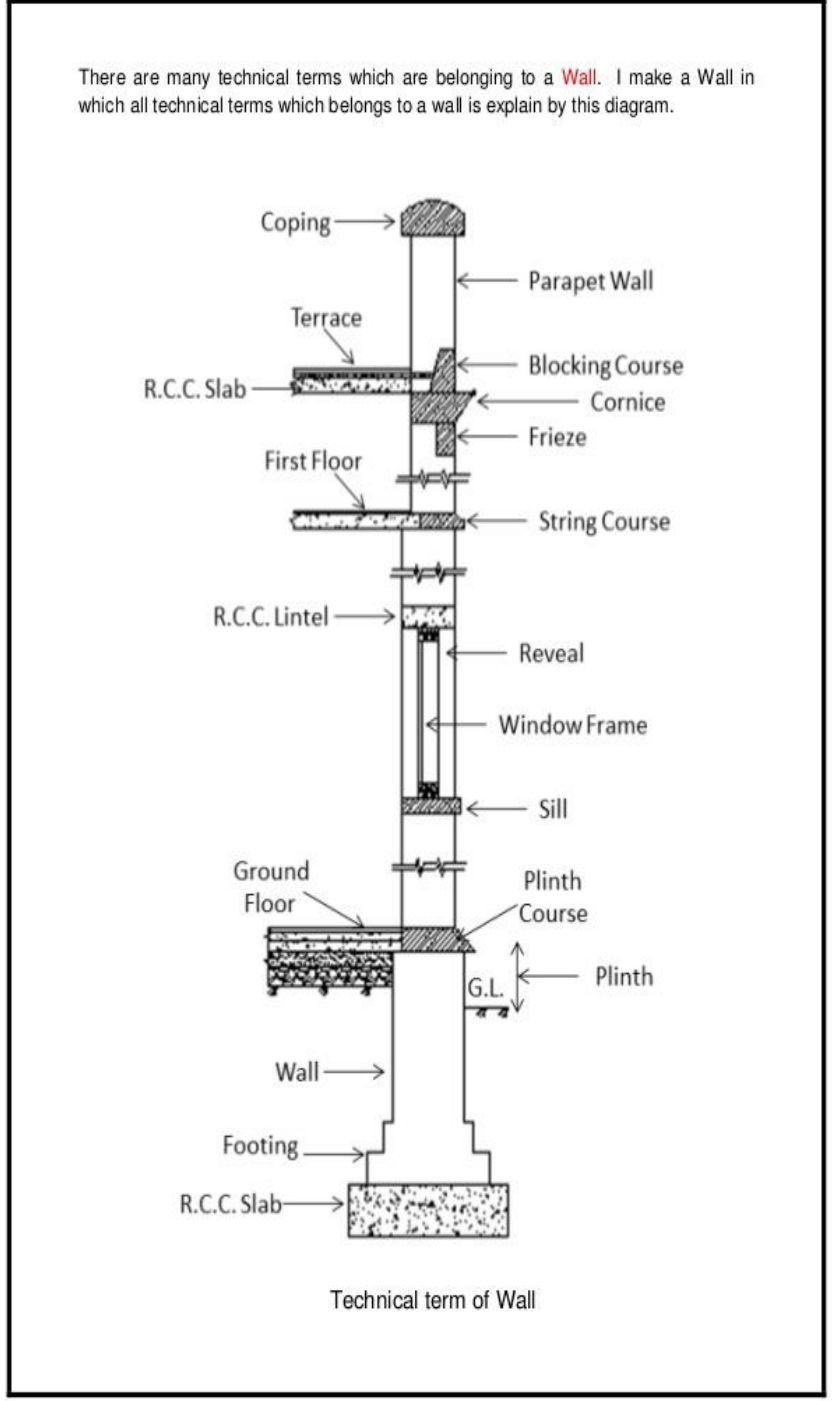
- Dead Loads:

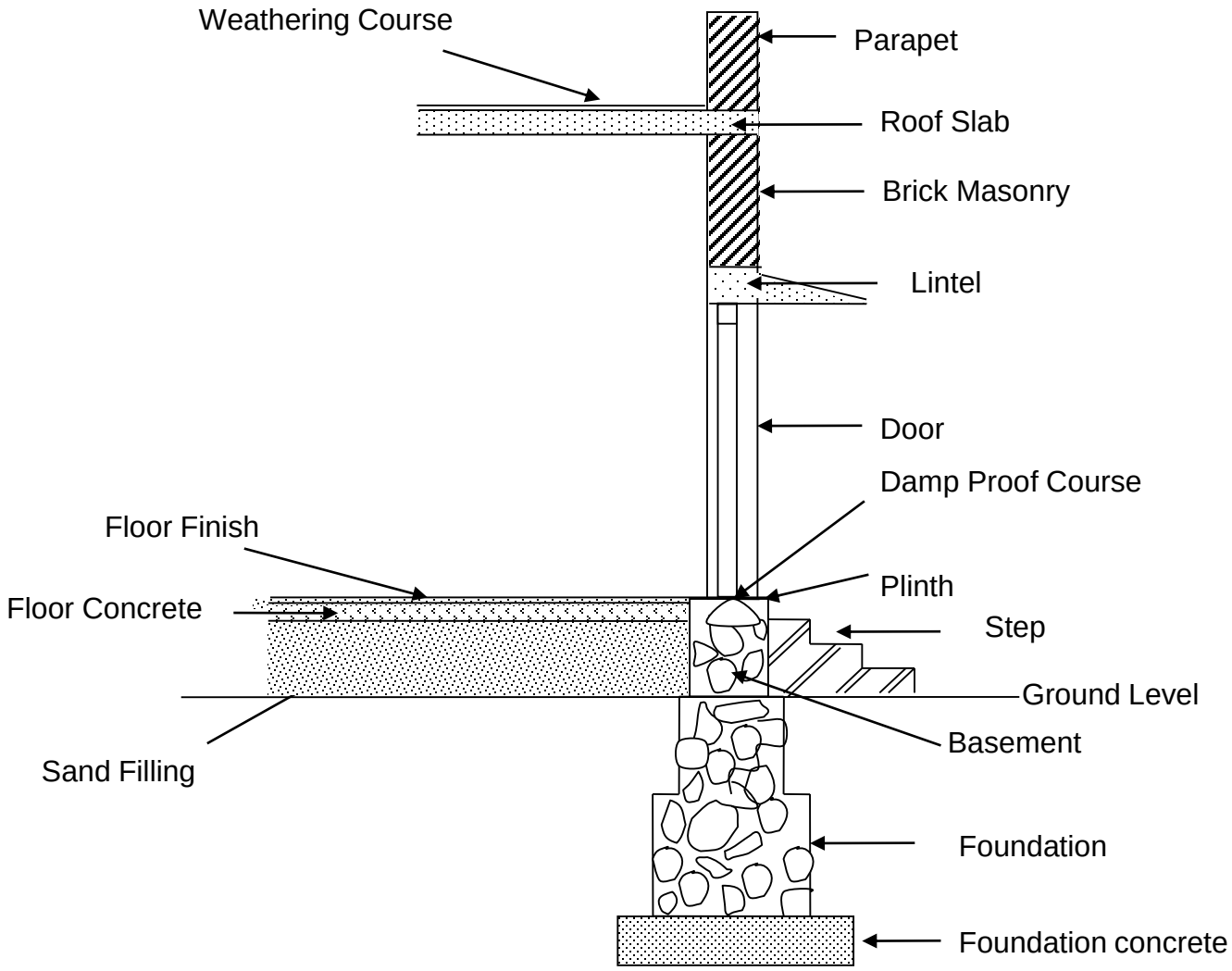
  "*Dead loads* consist of the weights of the various structural members and the weights of any objects that are permanently attached to the structure."

- Live Loads:
  - ✓ Occupancy Loads
  - ✓ Wind Loads
  - ✓ Snow Loads
  - ✓ Earthquake Loads
  - ✓ Other Loads:
    - ❑ Blast Loads
    - ❑ Variance in temperatures
    - ❑ Uneven settling of soil

Components of Buildings

- Any building has two parts
  - superstructure
  - substructure
- The various components of the superstructure are
  - Masonry units such as walls and columns
  - Floors
  - Doors, windows and other openings
  - Roof structures
  - Elevators, ramps and
  - Building finishes





COMPONENTS OF A BUILDING

Components of framed building structure



This building has ground floor, first floor, second floor, and terrace floor.

The vertical elements are the **columns.**

The horizontal bands are the **beams.**

The flat surface on which you can stand is the **slab.**

Walls, windows are added later to give protection to inhabitants.

The loads such as human beings, furniture etc. is carried by this frame.

The walls have no role except protecting the inhabitants from weather.

Components of framed building structure



This is a completed building as a framed building.

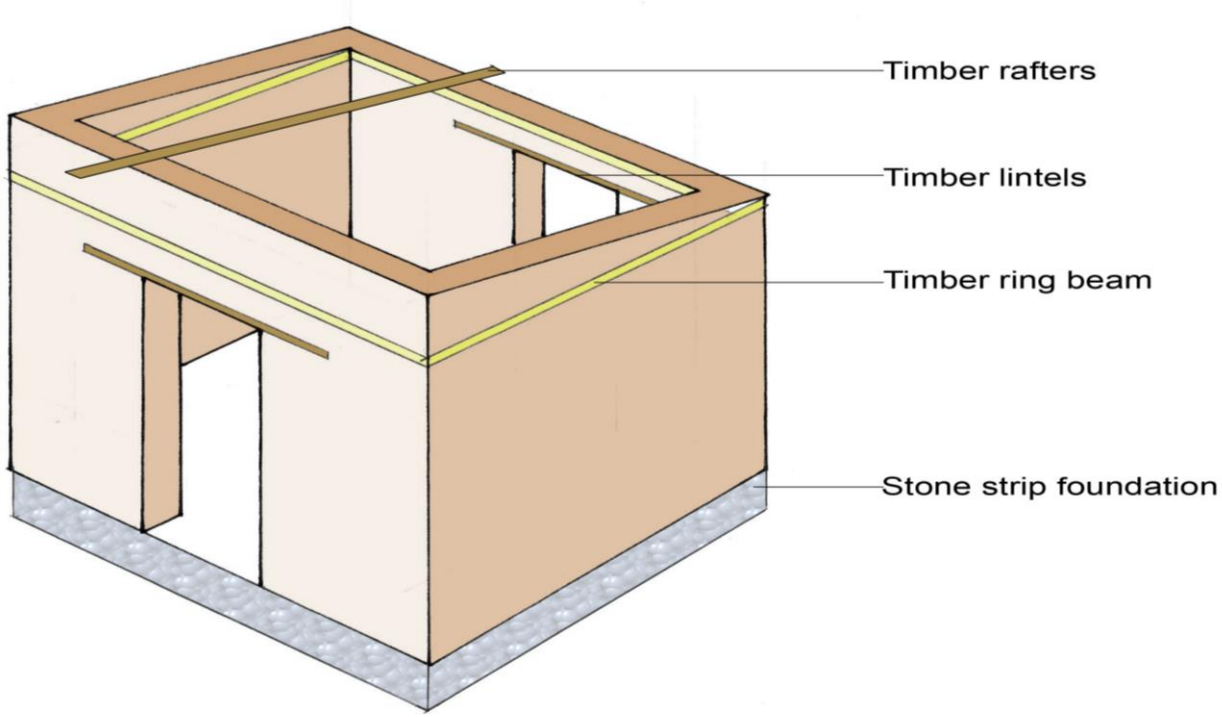
From the photo, you can see the red color; brick walls, grey color; - vertical columns and beams.

Framed building

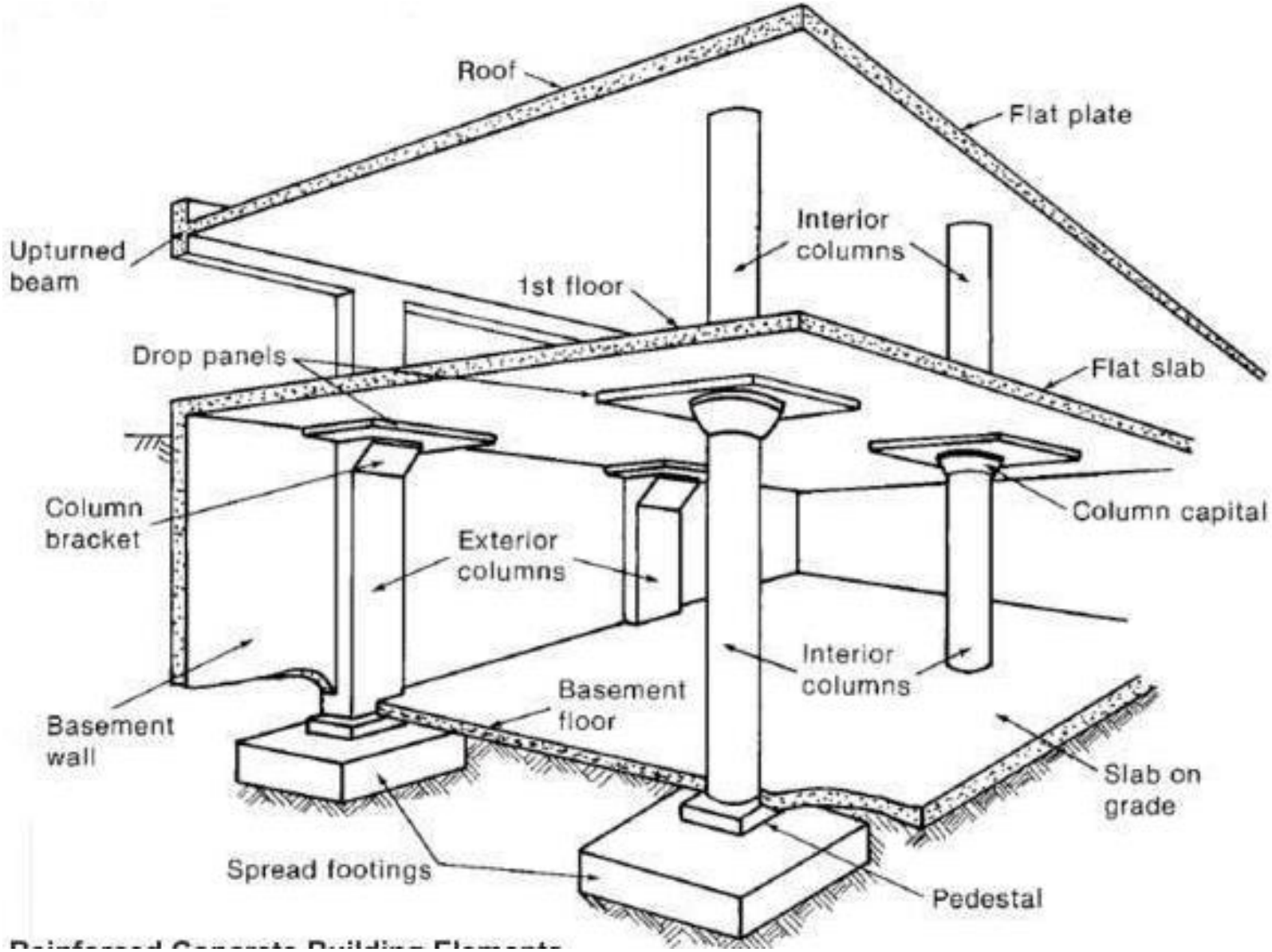


Load Bearing Structure

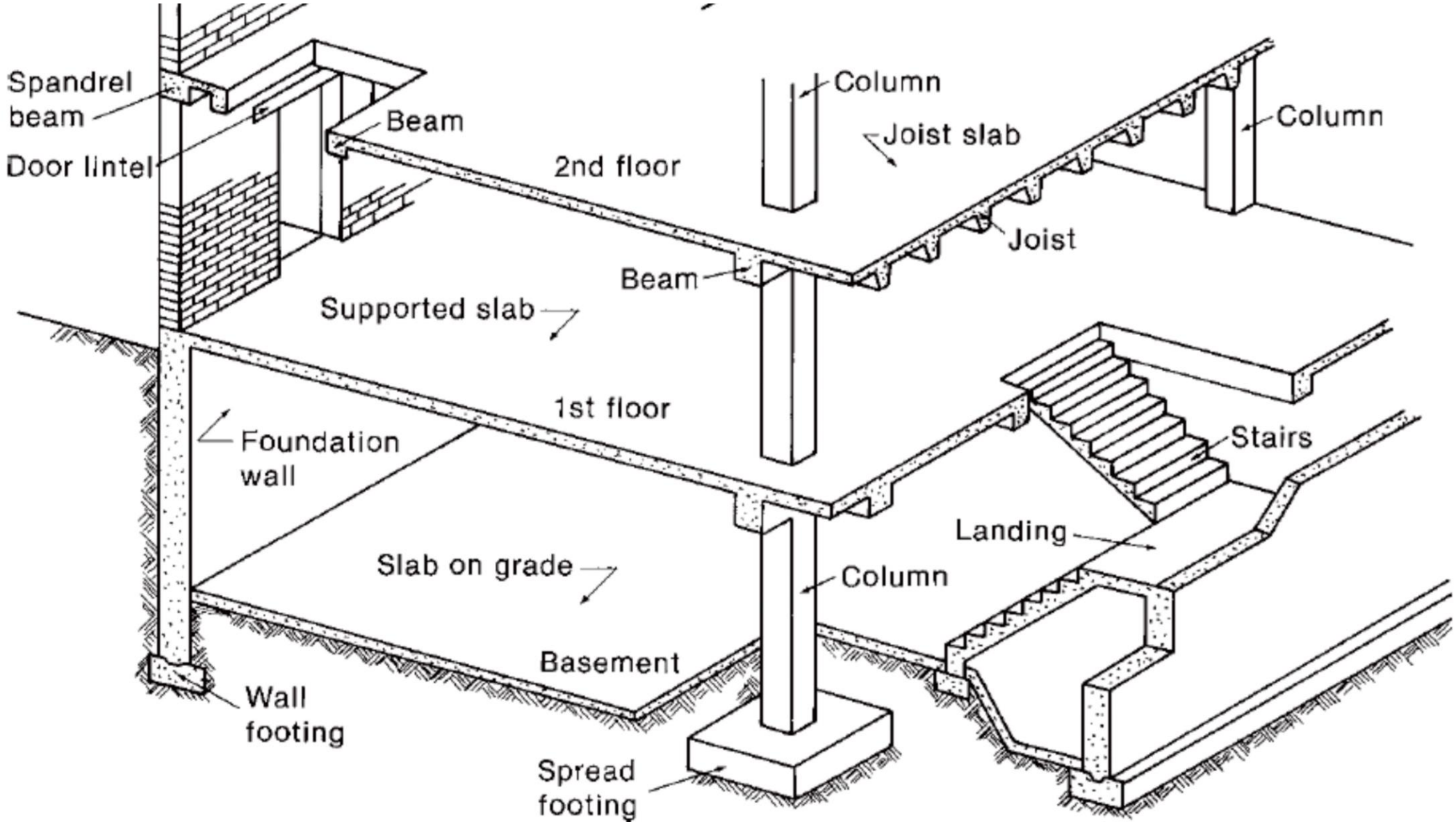
- Instead of columns and beams, it has walls taking the load.
- Walls play a dual role of taking loads and protecting the inhabitants. The walls are generally brick or stone.
- Since the walls take the load they cannot be moved or removed.
- The older structures of 2 to 3 storey were load bearing.
- Damaged wall can result in collapse



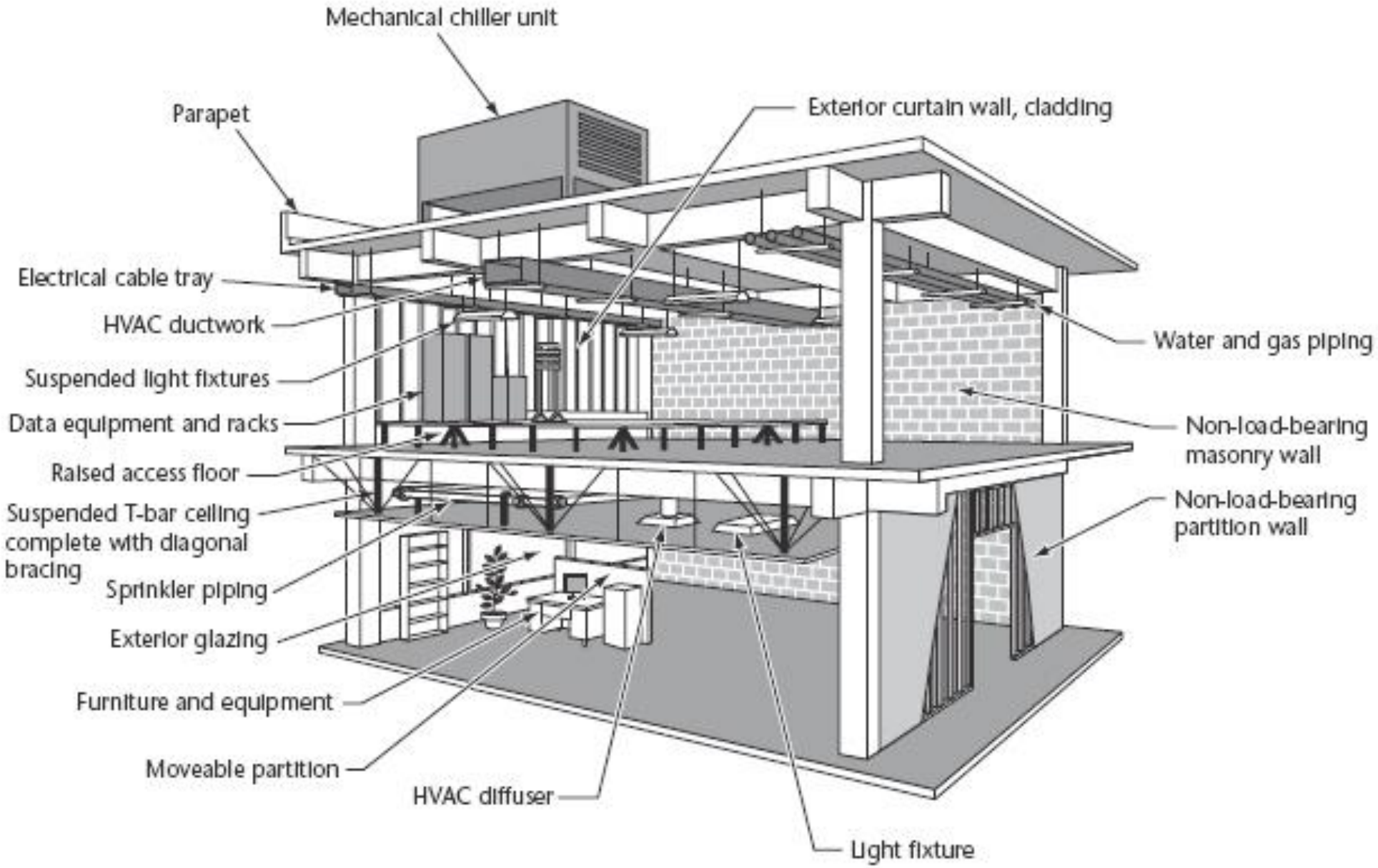
Structural components of reinforced concrete buildings



Structural components of reinforced concrete buildings



services for Buildings



FOOTING



Distribute load of structure to wide surface area to prevent any type of movement



Plinth

Dividing line between the substructure and superstructure

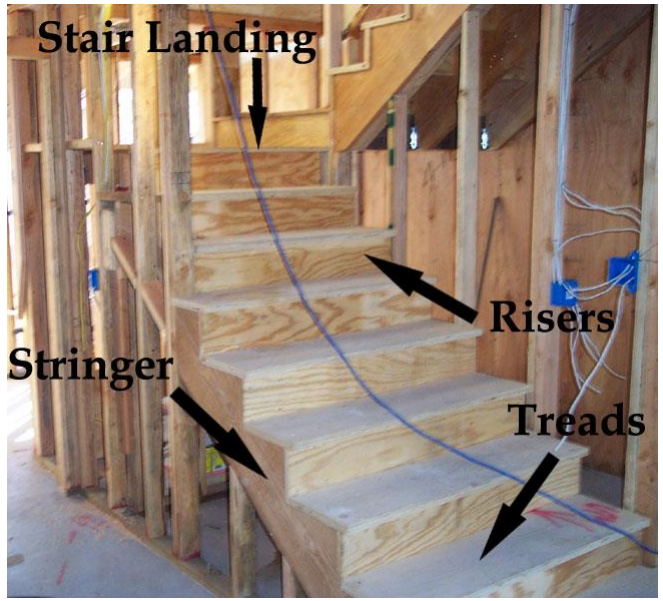






Stairs, Lifts, Escalators











WALLS

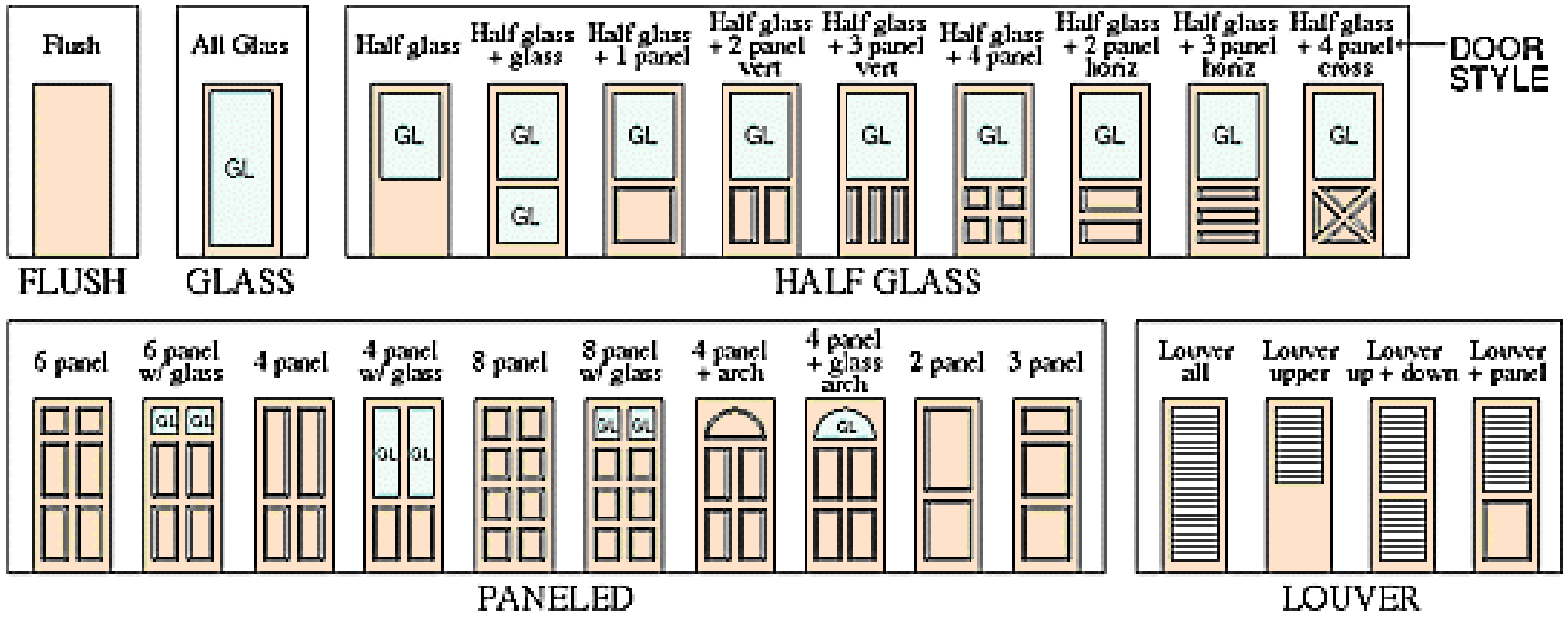
- It is the vertical structural members which may support to the roof or act as partition wall or compound wall.

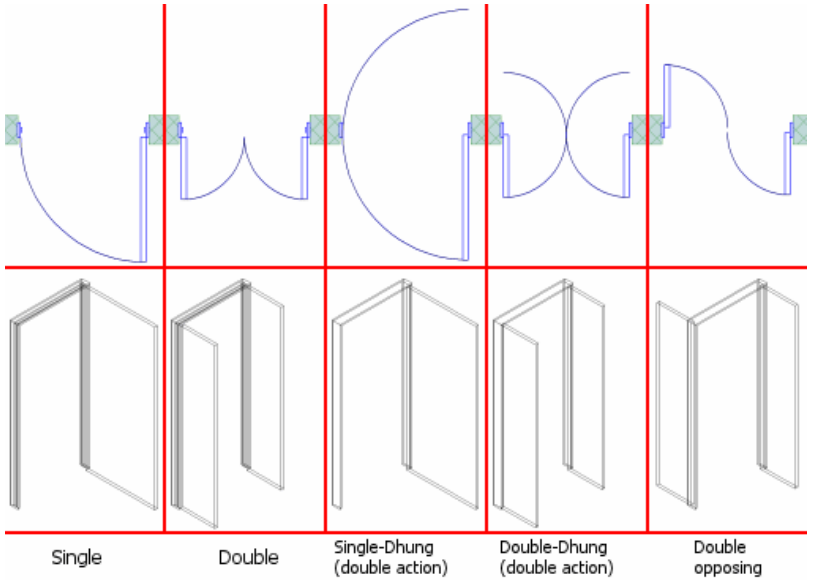
- Walls in various forms of masonry work such as brick masonry, stone masonry, composite masonry, hollow bricks.



Doors

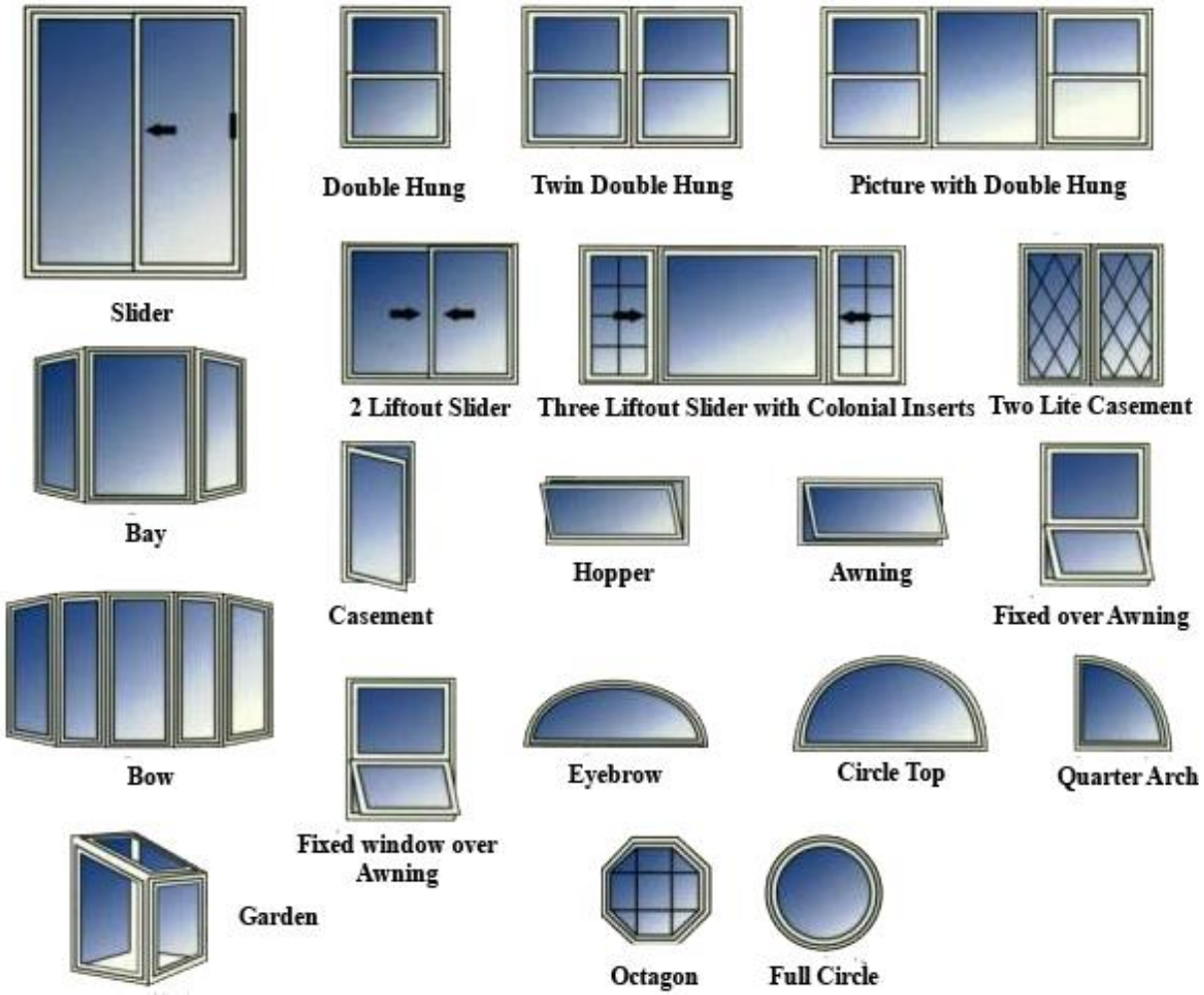
These are the openings which allow entrance in the building and circulation through different rooms.





Windows

- These are opening generally constructed in the external wall which provides air and light inside the rooms .

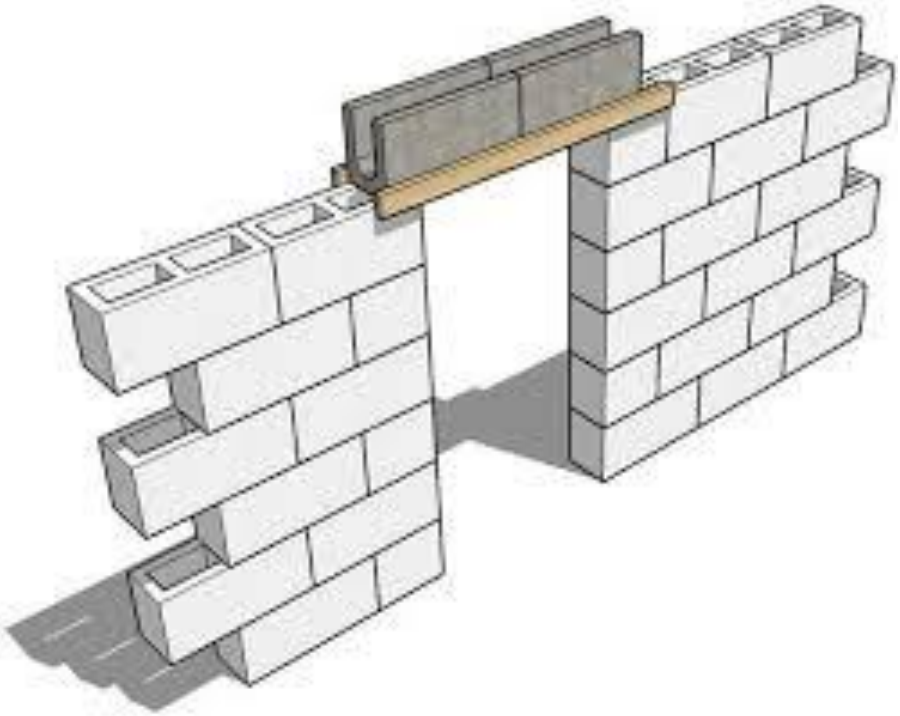




Lintels

- A beam that carries the wall load lies above doors and windows



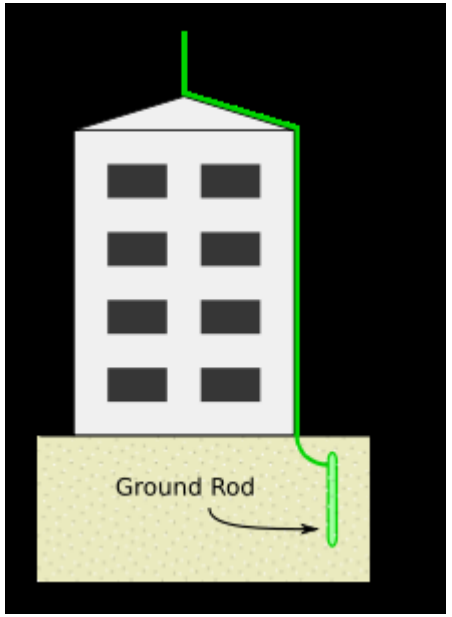


Building services

- Electrical works
  - ✓ Conduit (concealed / open) & Wiring
  - ✓ Cable Tray / Cable laying
  - ✓ Transformer / Electrical Panel / Diesel generators (Cooling tower) /Distribution board / UPS
  - ✓ Earthing / Lightning protection
  - ✓ Light fixtures / Power sockets











Building services

- Plumbing services
  - ✓ Pipe laying (Water / sewage)
  - ✓ Pumps / Tanks / Sump
  - ✓ Sewage treatment plants
  - ✓ Sanitary fittings









Building services

- Fire fighting system
  - ✓ Pipe laying
  - ✓ Pumps
  - ✓ Sprinkler system
  - ✓ Automatic fire detection system (fire alarm system)
  - ✓ Pipe laying, cable laying, detectors, hooters…



Fire fighting system



Smoke detectors



Sprinkler systems





Building services

- Air conditioning /ventilation
- -duct laying
- -AHU (Air handling units), chillers and pipes, sensors, heat Insulation
- -Exhaust fans

CHILLER UNIT



AHU



HVAC DUCT



Dispensers



Other services



- BMS (Building Management System)
- Access control
- CCTV Cameras



- PA (Public address / BGM (Back ground music system/ETBS (Emergency talk back system)

Door access control

Boom barrier



